

Introduction into Covariate Shift

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1 Introduction

Covariate shift in statistical learning refers to a situation where the distribution of the input features (covariates) changes between the training phase and the testing phase, while the conditional distribution of the output given the input remains the same¹. This means that while the relationship between the input features and the target variable doesn't change, the way the input features are distributed does.

To break this down:

- **Input Distribution Change:** The probability distribution $\mathbb{P}(X)$ of the input variables X changes between training and testing.
- **Conditional Distribution Stability:** The conditional distribution $\mathbb{P}(Y | X)$ of the target variable Y given the input variables X remains the same.

In mathematical terms, if during training we have a distribution $\mathbb{P}_{\text{train}}(X, Y)$, and during testing, we have a distribution $\mathbb{P}_{\text{test}}(X, Y)$, covariate shift means:

$$\mathbb{P}_{\text{train}}(X) \neq \mathbb{P}_{\text{test}}(X)$$

$$\mathbb{P}_{\text{train}}(Y | X) = \mathbb{P}_{\text{test}}(Y | X) .$$

Covariate shift can lead to poor model performance on the test data because the model was trained on a different distribution of input features. The model may not generalize well if the training data does not adequately represent the test data.

¹see e.g. <https://www.doc.ic.ac.uk/~bkainz/teaching/DL/T04covariateShift.pdf>

There are several techniques to handle covariate shift:

1. **Reweighting Samples:** Adjust the importance of training samples to better match the test distribution. This can be done using techniques like importance sampling, where each training sample is weighted by the ratio of the test distribution to the training distribution.
2. **Domain Adaptation:** Adapt the model to perform well on the test distribution by using techniques specifically designed for learning under distributional changes.
3. **Data Augmentation:** Generate additional training data that is more representative of the test distribution, if possible.
4. **Transfer Learning:** Leverage models or features learned from different but related tasks where the distributions might be similar to the test distribution.
5. **Robust Models:** Develop models that are inherently robust to distributional changes, though this can be challenging and context-dependent.

Example. Imagine a spam email classifier trained on emails from 2020. The types of emails and spam tactics may change by 2023, leading to a different distribution of email content. If the classifier does not account for this shift, its performance could degrade. However, the way emails are classified as spam (based on features like keywords, sender reputation, etc.) might remain the same, indicating a covariate shift rather than a change in the underlying spam classification logic.

Understanding and mitigating covariate shift is crucial for building robust and reliable machine learning models, especially in dynamic environments where data distributions are prone to change over time.

2 Supervised Learning under Covariate Shift

The goal of supervised learning (regression or classification) is to predict a real-valued response Y based on a vector of covariates X , being a random variable with values in some input space $\mathcal{X}$. We assume that the pair (X, Y) has unknown distribution $\mathbb{P}$ and X has marginal $\mathbb{P}_X$. We refer to $\mathbb{P}_X$ as the *source distribution*. The aim is to minimize the risk (loss, cost)

$$f^* \in \arg \min_f \mathcal{R}(f), \quad \mathcal{R}_{\mathbb{P}}(f) := \mathbb{E}[l(Y, f(X))],$$

where the expectation is taken with respect to the distribution $\mathbb{P}$. Here, l denotes a loss function.

Example 2.1 (Regression). *For each fixed $x \in \mathcal{X}$, the optimal estimator in a mean-squared sense is given by the regression function $f^*(x) := E[Y|X = x]$, that is,*

$$f^* \in \arg \min_{f \in L^2(\mathcal{X}, \mathbb{P}_X)} \mathcal{R}(f), \quad \mathcal{R}_{\mathbb{P}}(f) := \mathbb{E}[(Y - f(X))^2],$$

A standard calculation shows that for all $f \in L^2(\mathcal{X}, \mathbb{P}_X)$, the excess risk is given by

$$\mathcal{R}_{\mathbb{P}}(f) - \mathcal{R}_{\mathbb{P}}(f^*) = \|f - f^*\|_{L^2(\mathbb{P}_X)}^2.$$

To find an estimator $\hat{f}_D$ for f^* , we are given an i.i.d. training sample $D = ((X_1, Y_1), \dots, (X_n, Y_n)) \in (\mathcal{X} \times \mathcal{Y})^n$, i.e. $D \sim \mathbb{P}^n$.

In the covariate shift version of this problem, the *target distribution* $\mathbb{Q}_X$ is different from the source distribution $\mathbb{P}_X$. The main goal is to find an estimator $\hat{f}_D$ trained on $D \sim \mathbb{P}^n$ whose excess risk with respect to $\mathbb{Q}_X$ is small, i.e.

$$\mathcal{R}_{\mathbb{Q}}(\hat{f}_D) - \mathcal{R}_{\mathbb{Q}}(f^*) = \|\hat{f}_D - f^*\|_{L^2(\mathbb{Q}_X)}^2$$

is small.

It may be useful to assume that $\mathbb{P}_X$ is in some sense "close to" $\mathbb{Q}_X$.

Simplifying assumptions: We assume there exists a function $w : \mathcal{X} \rightarrow \mathbb{R}_+$ such that

$$d\mathbb{Q}_X(x) = w(x)d\mathbb{P}_X(x)$$

and w is the Radon-Nikodym derivative of the target measure with respect to the source measure.

Assume either

1. $\sup_{x \in X} |w(x)| \leq B$,i.e. uniformly bounded
2. $\mathbb{E}_{\mathbb{P}_X}[w(X)^2] \leq \tau^2, \tau^2 \geq 1$.

Assumption 1 is the more strict condition (and not satisfied for some classical distributions since $\mathbb{E}_{\mathbb{P}}[w(X)^2] = \mathbb{E}_{\mathbb{Q}}[w(X)] \leq \tau^2$ and so the uniform bound B includes the second moment bound $\tau^2 = B$).

It may be useful to choose B later with the sample size of the training sample, as in [Ma et al. \(2022\)](#).

More precisely, we may define for some $\alpha > 0$ the truncated density

$$\omega_{\alpha}(x) := 1_{\{\omega(x) \leq \alpha\}}(x) \cdot \omega(x) .$$

Note that this implies that $\omega_{\alpha}(x) \leq \alpha$ for all $x \in \mathcal{X}$ and allows to tackle the unbounded case.

Note that the second assumption implies that the χ^2 -distance between the marginals is bounded as well, since

$$\chi^2(\mathbb{P}_X || \mathbb{Q}_X) = \mathbb{E}_{\mathbb{P}_X}[\omega^2(X)] - 1 .$$

Example 2.2 (Mean shift). *As an example we consider a mean shift in the family of the multivariate normal distributions:*

$$\mathbb{P}_X = \mathcal{N}(0, \Sigma) , \quad \mathbb{Q}_X = \mathcal{N}(\mu, \Sigma) ,$$

where $\mu \in \mathbb{R}^d$ is the mean and positive definite $\Sigma : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is the covariance matrix. Recall that the density of the normal distribution is given by

$$f_{\mu, \Sigma}(x) = (2\pi)^{-d/2} \det(\Sigma)^{-1/2} \exp\left(-\frac{1}{2}(x - \mu)^T \Sigma^{-1/2}(x - \mu)\right) .$$

Hence, the weights can be calculated directly and are given by

$$\omega(x) = e^{\frac{1}{2}(\mu^T \Sigma^{-1} \mu + 2 \cdot x^T \Sigma^{-1} \mu)} .$$

However, they are not bounded!

3 Covariate shift in Linear Regression

We focus on linear regression, i.e. $\mathcal{X} = \mathbb{R}^d$, $\mathcal{Y} = \mathbb{R}$, and our observations follow the model

$$y_j = f^*(x_j) + \varepsilon_j, \quad f^*(x_j) = \langle \theta^*, x_j \rangle, \quad (1)$$

with $j = 1, \dots, n$, $\theta^* \in \mathbb{R}^d$. The training data $D = ((x_j, y_j))_j$ are i.i.d. from $\mathbb{P}$. The noise satisfies $\varepsilon_j \sim \mathcal{N}(0, \sigma^2)$. We denote the design matrix by $\mathbf{X} \in \mathbb{R}^{n \times d}$ and the vector of outputs by $\mathbf{Y} = (y_1, \dots, y_n)^T \in \mathbb{R}^n$. We also assume we have a set of test data that are i.i.d. with respect to the target distribution: $(\tilde{x}_j, \tilde{y}_j) \sim \mathbb{Q}$, $j = 1, \dots, m$.

3.1 Importance Weighted Regularization

In the following, we analyze the performance of different regularization methods for estimating θ^* under covariate shift using the method of *importance weighting*. To this end, let $M_\omega : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be the diagonal matrix with diagonal entries $(\omega(x_1), \dots, \omega(x_n))$, i.e.,

$$M_\omega = \begin{bmatrix} \omega(x_1) & 0 & \cdots & 0 \\ 0 & \omega(x_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \omega(x_n) \end{bmatrix}.$$

OLS. The *importance weighted ordinary least-squares estimator* is given by solving the empirical risk minimization problem on the training data

$$\hat{\theta}_{OLS} := \arg \min_{\theta \in \mathbb{R}^d} \frac{1}{n} \sum_{j=1}^n \omega(x_j) \left(\langle \hat{\theta}, x_j \rangle - y_j \right)^2.$$

The solution to that problem is given by

$$\hat{\theta}_{OLS} = (\mathbf{X}^T M_\omega \mathbf{X})^{-1} \mathbf{X}^T M_\omega \mathbf{Y}. \quad (2)$$

Ridge regression. The *importance weighted ridge regression estimator* is given by solving the regularized empirical risk minimization problem on the training data

$$\hat{\theta}_{RR} := \arg \min_{\theta \in \mathbb{R}^d} \frac{1}{n} \sum_{j=1}^n \omega(x_j) \left(\langle \hat{\theta}, x_j \rangle - y_j \right)^2 + \lambda \|\theta\|_2^2,$$

with $\lambda > 0$ being the regularization parameter. The solution to that problem is given by

$$\hat{\theta}_{RR} = (\mathbf{X}^T M_\omega \mathbf{X} + \lambda Id_d)^{-1} \mathbf{X}^T M_\omega \mathbf{Y}. \quad (3)$$

Gradient Descent. Similarly as for OLS and RR, we introduce the importance weighted gradient descent algorithm: $\hat{\theta}_0 = 0$ and for $t = 1, \dots, T-1$, the iterates are recursively defined as

$$\hat{\theta}_{t+1} := (Id_d - \gamma \cdot \mathbf{X}^T M_\omega \mathbf{X}) \hat{\theta}_t + \frac{\gamma}{n} \mathbf{X}^T M_\omega \mathbf{Y}.$$

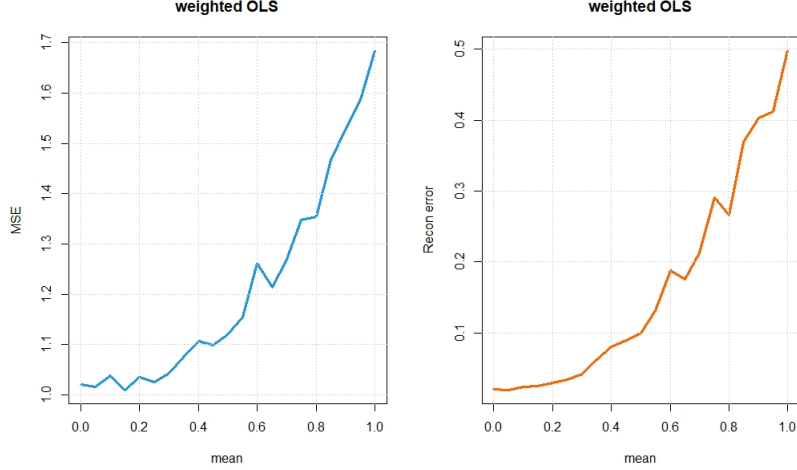


Figure 1: Error in the setting of mean shift under normally distributed data for the OLS estimator. We observe a lower performance under distribution shift. **Left:** Empirical risk on target data. **Right:** Reconstruction error.

Here, $\gamma < 1/\|\mathbf{X}^T M_\omega \mathbf{X}\|$ is the positive constant stepsize and needs to be sufficiently small. The number $T > 0$ is the stopping time. Note that the operator norm $\|\mathbf{X}^T M_\omega \mathbf{X}\|$ is equal to the largest eigenvalue of the matrix $\mathbf{X}^T M_\omega \mathbf{X}$.

Note that in practical applications, we do not know the weights as we do not know the underlying data generating process (the probability distributions). We need a strategy to estimate or learn them from the data by e.g.:

- density ratio estimation, or
- learning them along with with some gradient based algorithm (consider the as trainable parameters), called *self-attention*.

Measure of performance. For an estimator $\hat{f}(x) = \langle \hat{\theta}, x \rangle$, we measure the performance in two ways:

- **Norm error or Reconstruction error:** $\|\hat{\theta} - \theta^*\|_2$.
- **Empirical risk on target/ test data:** $\frac{1}{m} \sum_{j=1}^m (\hat{f}(\tilde{x}_j) - \tilde{y}_j)^2 = \frac{1}{m} \sum_{j=1}^m \left(\langle \hat{\theta}, \tilde{x}_j \rangle - \tilde{y}_j \right)^2$.

3.2 Self-attention: Learning the weights with Gradient Descent

Since we do not know the weights we introduce the strategy of self-attention and want to learn the weights along with gradient descent and perform a gradient descent-ascent algorithm: minimizing the empirical loss with respect to the unknown regression parameters and maximizing the empirical loss w.r.t. the weights. Let the empirical loss be given by

$$\hat{\mathcal{R}}(\theta, w) := \frac{1}{n} \sum_{j=1}^n m(w^{(j)}) l(y_j, \langle \theta, x_j \rangle),$$

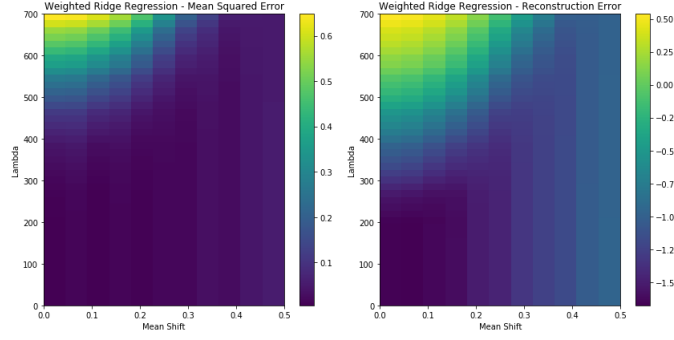


Figure 2: Error in the setting of mean shift under normally distributed data for Ridge Regression. With increasing mean shift, the optimal regularization parameter increases.

with $w = (w^{(1)}, \dots, w^{(n)})$ and $m : \mathbb{R} \rightarrow \mathbb{R}$ being a *masking function*. The gradient descent-ascent algorithm is given by: $\theta_0 = 0$, $w_0^{(j)} = 0$ for all $j = 1, \dots, n$ and

$$\begin{aligned}\theta_{t+1} &= \theta_t - \gamma_t \nabla_{\theta} \hat{\mathcal{R}}(\theta_t, w_t) \\ w_{t+1} &= w_t + \eta_t \nabla_w \hat{\mathcal{R}}(\theta_t, w_t).\end{aligned}$$

Here, $(\gamma_t)_t$ and $(\eta_t)_t$ are the learning rates.

Example 3.1 (Masking function). *Possible choices are: $m_q(t) = t^q$ for some $q \geq 0$ (unbounded) or $m(t) = \tanh(t)$ (bounded).*

The self-attention algorithm has the advantage that we do not need additional data to learn the weights.

4 Covariate Shift Regularization with Kernels

Project outline. The main goal of this project is to understand the impact of covariate shift on learning algorithms and performance.

- Rephrase the covariate shift setting in the framework of kernel methods, see e.g. [Mohri \(2018\)](#), [Bach \(2024\)](#). What is covariate shift and which strategies are used to tackle this?
- Summarize the literature and state of the art, see e.g. [Gogolashvili et al. \(2023\)](#), [Ma et al. \(2022\)](#). You find also an experiment there for some inspiration.
- Implement at least two methods: One method (the baseline) by estimating the weight function using a method for function estimating (e.g. density ratio estimation) combined with stochastic gradient descent and/or g gradient descent and the second using the self-attention mechanism. Try also a stochastic version. Compare the methods with no importance weighting.
- Compare the algorithms with respect to complexity: Early stopping, choice of learning rate, sample sizes of both the training and test data sets, masking function,
- If time allows: Can we do a low-rank approximation of the kernel matrix and maintain efficiency?

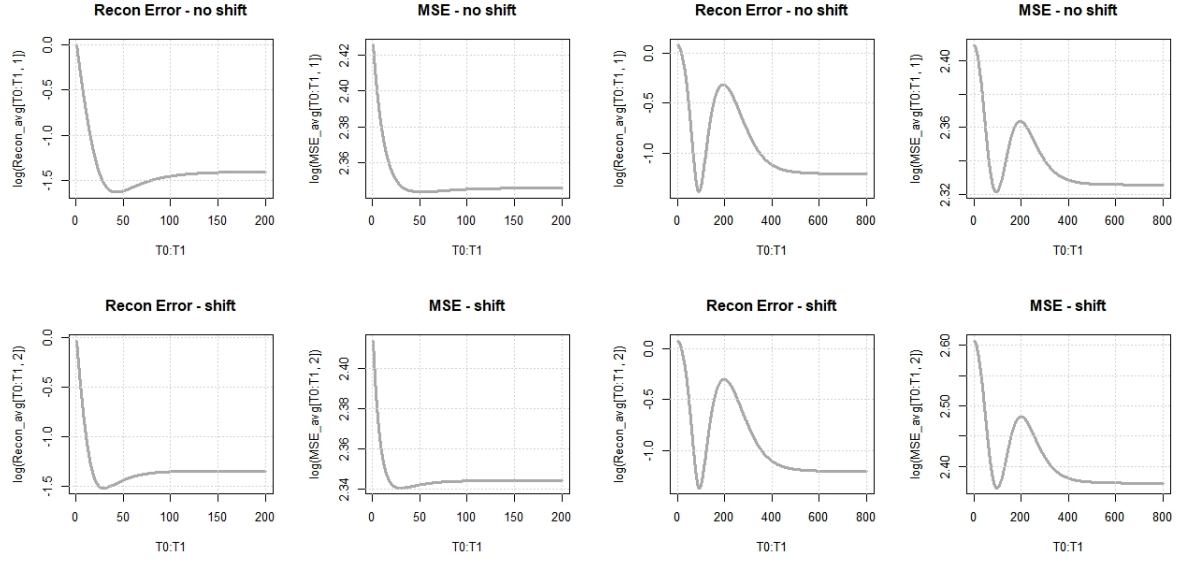


Figure 3: Reconstruction error and mean squared error versus number of iterations in the setting of mean shift under normally distributed data. We observe a local minimum and an optimal early stopping time. **Left:** Empirical risk for Gradient Descent with no shift and mean shift with known weights. **Right:** Gradient Descent-Ascent with masking function $m(t) = t$. The weights are learned.

References

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